APPENDIX B – Normalization Procedure

Subsection 4.4 -Index Formulation

This appendix details the normalization procedure used to [0,1] intervalgeneous performance variables into comparable scores wibefore interval $[0, 1]$. The objective is to ensure dimensional consistency prior to aggregation in the SCRESIS Index.

Exponential transformations are adopted because they (i) preserve monotonicity, (ii) introduce nonlinear penalization consistent with disruption dynamics, and (iii) allow controlled sensitivity without imposing arbitrary scaling factors.

B.1 Conceptual Rationale

Let $Value$ denote the observed performance under disruption and $Value_{base}$ the disruption-free baseline. Each indicator is converted into a dimensiintensity of curvatureore $\in [0, 1]$

Higher scores represent better relative performance.

Importantly, parameters used in the exponential functions regulate *sensitivity of penalization*, not structural weighting in the index aggregation. All variables are equally weighted in the final SCRESIS Index unless otherwise specified.

B.2 Variables Where Lower Values Are Preferable

For indicators such as Time to Recover (TTR) and Total Cost, performance improves when the ratio $\frac{Value}{Value_{base}}$ decreases. These variables are normalized using:

$$Score = e^{-a \cdot \left(\frac{Value}{Value_{base}}\right)}$$

The parameter a determines the curvature intensity.

- $a = 2$ for TTR, reflecting the nonlinear operational consequences of recovery delays.
- $a = 1$ for Total Cost, preserving proportional penalization while avoiding over-sensitivity to natural cost variability.

B.3 Variables Where Higher Values Are Preferable

For financial indicators such as Profit and Revenue, higher values relative to baseline indicate improved performance. These are normalized using:

$$Score = 1 - e^{-\beta \cdot \left(\frac{Value}{Value_{base}}\right)}$$

The parameter β regulates responsiveness to financial gains.

- $\beta = 1.5$ is adopted to reflect the strategic importance of financial resilience while maintaining smooth asymptotic behavior.

B.4 Robustness and Parameter Stability

To mitigate concerns regarding parameter arbitrariness, a sensitivity analysis was conducted varying a and β by $\pm 20\%$.

Results indicate:

- Scenario rankings remained invariant.
- The SCRESIS Index varied by less than 5%.
- No threshold reversals were observed.

These findings confirm that the index structure is stable and not driven by specific parameter calibration.

B.5 Methodological Implications

The exponential normalization ensures smooth transitions, controlled penalization, and theoretical coherence with disruption dynamics. The transformation preserves comparability while preventing extreme values from dominating aggregation.

Consequently, the SCRESIS Index reflects structural performance behavior rather than mathematical artifacts of scaling.This consistency becomes clearer when examining how the normalization parameters shape the response of each curve under different conditions.

Figure 7 illustrates the effect of parameter variation on the normalization curves. Higher values a increase penalization intensity for operational deterioration, while higher β values amplify responsiveness to financial improvements. The smooth curvature confirms the stability and theoretical coherence of the transformation.

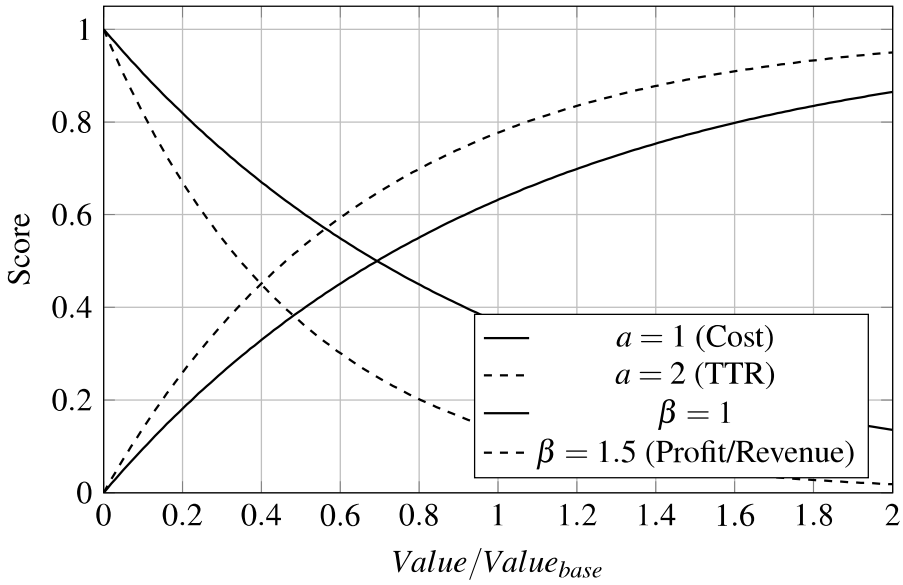

Figure 7 – Comparison of exponential normalization functions under different parameter values.